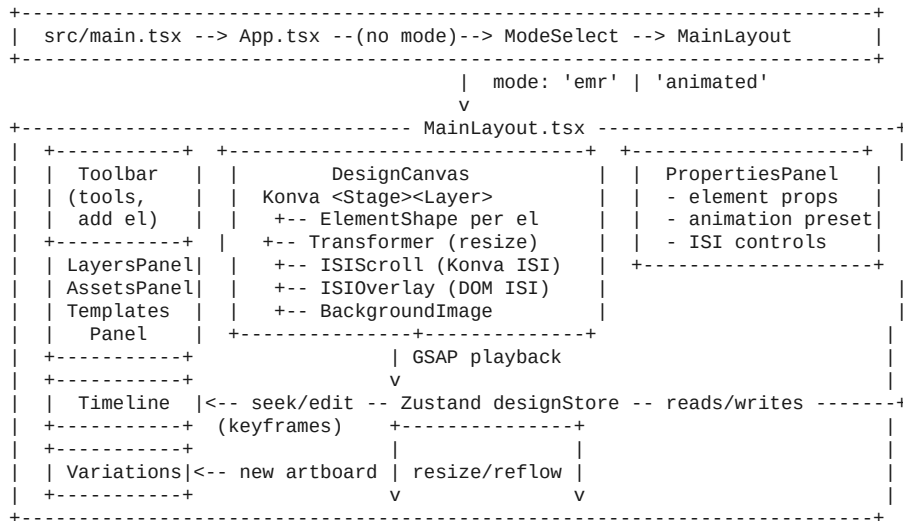


Animated Banner Tool — Build Architecture

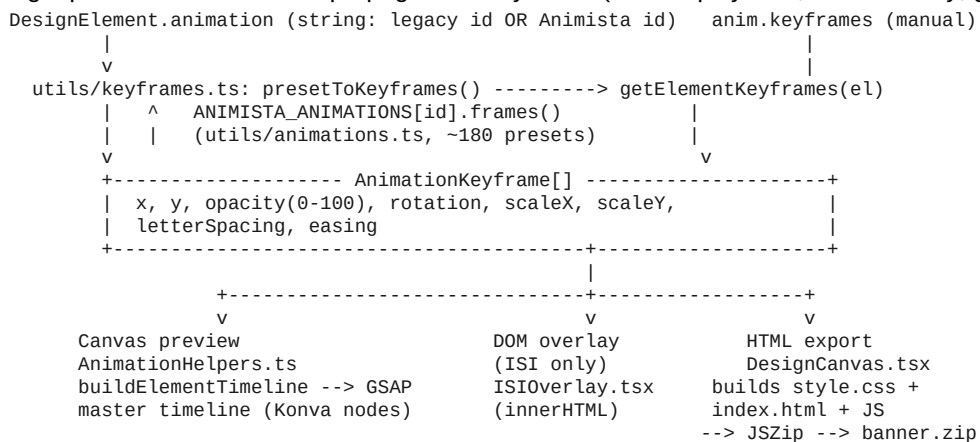
Stack: React 19 + TypeScript 5.9 + Vite 7 · Zustand 5 (state) · Konva/react-konva (canvas) · GSAP 3 (animations) · JSZip (HTML export) · Tailwind CSS · Vitest + ESLint.

1. High-level flow

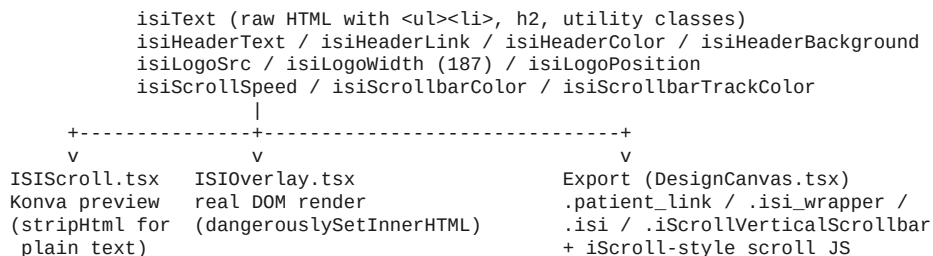


2. Animation / keyframe pipeline (single source of truth)

Every preview, overlay, and export path resolves keyframes through `getElementKeyframes(el) → presetToKeyframes(el)`. Registering a preset in `animations.ts` propagates everywhere (canvas playback, DOM overlay, generated banner JS).



3. ISI sub-system (dual-rendered)



4. State model (src/store/designStore.ts)

```
Artboard {id, label, width, height}
  ^  addArtboard() / resize / reflow
  |
  +-- elements: DesignElement[]

DesignElement (single polymorphic model)
  |
  +-- type: text|rect|circle|line|image|
  |     isiScroll|shape|html
  +-- transform/style props (x,y,fill,font...)
  +-- animation?: string (preset id)
  +-- anim?: { keyframes, loop } (manual)
  +-- animationDuration / animationDelay / animationLoop
  +-- isi* props (content, header, logo, scrollbar)
  +-- hover: hoverAnimation / hoverColor
```

5. Key decisions

- Single source of truth: animation presets live in utils/animations.ts and flow through keyframes.ts.
- Keyframe model supports x/y/opacity/rotation/scaleX/scaleY/letterSpacing; Animista blur/skew/clip-path are approximated.
- ISI is rendered twice: Konva (editable canvas) + real HTML (ISIOverlay) for fidelity; export emits the traditional .patient_link/.isi/.iScrollVerticalScrollbar markup.
- Generated banner is dependency-light: GSAP from CDN only (no jQuery/iScroll).
- Tests (34, Vitest) cover keyframe generation and store logic.

6. Mermaid version (for GitHub/Notion/VS Code)

```
graph TD
    App -->|no mode| ModeSelect --> MainLayout
    subgraph MainLayout ["MainLayout"]
        Toolbar & LayersPanel & AssetsPanel & TemplatesPanel
        subgraph Canvas ["DesignCanvas"]
            Stage[Konva Stage] --> Elem[ElementShape per element]
            Stage --> ISI[ISI Scroll Konva preview]
            Stage --> Ovl[ISIOverlay DOM preview]
        end
        end
        PropertiesPanel & Timeline & VariationsPanel
    end
    MainLayout <--> Store[Zustand designStore]
    Store -->|animation id| PF[presetToKeyframes]
    Anim[(animations.ts catalog)] --> PF
    Manual[(anim.keyframes)] --> GEK[getElementKeyframes]
    PF --> GEK --> KF[AnimationKeyframe[]]
    KF --> TH[AnimationHelpers buildMasterTimeline] --> Stage
    KF --> EXP[DesignCanvas export JS]
    EXP --> ZIP[JSZip HTML package]
```